FTP (File Transfer Protocol) Architecture

Coverage: Out-of-Band Control Port 21 vs In-Band Data Port 20, Active Mode (PORT) vs Passive Mode (PASV), Stateful Operation

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1. Dual-Channel Architecture

- Control Connection: Dedicated TCP port 21 for commands (USER, PASS, RETR, STOR) and status replies. Remains open.
- Data Connection: Dedicated TCP port 20 opened per file transfer and closed upon completion.
- Stateful: Server tracks current working directory, authenticated user, and transfer mode.

2. Active vs Passive Mode

- Active Mode: Client sends PORT command; Server initiates data connection from Port 20 to Client (blocked by client firewalls).
- Passive Mode: Client sends PASV command; Server opens random high port; Client initiates data connection (firewall-friendly).

■ KTU University High-Yield Exam Questions

- Q: Explain why FTP is called an out-of-band protocol. Compare Active and Passive FTP. (8 Marks)